28 Jan 2026 Shift 2 (Actual paper)

Math Sec 1

1. Let $f(x)=\int \frac{\mathrm{d}x}{x^{\left(\frac{2}{3}\right)}+2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0)=-26+24\log_e(2)$. If $f(1)=\mathrm{a}+\mathrm{b}\log_e(3)$, where a , b $\in \mathbf{Z}$, then $\mathrm{a}+\mathrm{b}$ is equal to :

(A) -26 (B) -18 (C) -5 (D) -11

2. Let P be a point in the plane of the vectors $\overrightarrow{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equidistant from the lines AB and AC. If $|\overrightarrow{AP}|=\frac{\sqrt{5}}{2}$, then the area of the triangle ABP is:

(A) $\frac{3}{2}$ (B) $\frac{\sqrt{26}}{4}$

(C) 2 (D) $\frac{\sqrt{30}}{4}$

3. Let [] denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\overline{2}}\left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right)\mathrm{d}x$ is equal to:

(A) $13\pi+1$ (B) $12\pi+5$ (C) $11\pi+2$ (D) $15\pi+4$

4. The probability distribution of a random variable X is given below :

X	4 k	$\frac{30}{7}$k	$\frac{32}{7}$k	$\frac{34}{7}$k	$\frac{36}{7}$k	$\frac{38}{7}$k	$\frac{40}{7}$k	6 k
$P(X)$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $\mathrm{E(X)}=\frac{263}{15}$, then $\mathrm{P(X<20)}$ is equal to :

(A) $\frac{14}{15}$ (B) $\frac{8}{15}$

(C) $\frac{11}{15}$ (D) $\frac{3}{5}$

5. Let $P_1: y=4x^2$ and $P_2: y=x^2+27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y=\alpha x, \alpha>0$ and P_1, then α is equal to :

(A) 15 (B) 12 (C) 8 (D) 6

6. Let $y=y(x)$ be the solution of the differential equation $x\frac{\mathrm{d}y}{\mathrm{d}x}-y=x^2\cot x, x\in(0,\pi)$. If $y\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right)-8y\left(\frac{\pi}{4}\right)$ is equal to :

(A) π (B) -3π (C) 3π (D) $-\pi$

7. Let $\mathrm{Q(a,b,c)}$ be the image of the point $\mathrm{P}(3,2,1)$ in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is

(A) 8 (B) 7 (C) 6 (D) 5

8. Let the ellipse $\mathrm{E}: \frac{x^2}{144}+\frac{y^2}{169}=1$ and the hyperbola $\mathrm{H}: \frac{x^2}{16}-\frac{y^2}{\lambda^2}=-1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H , then the value of $24(\mathrm{e}+\mathrm{L})$ is :

(A) 148 (B) 126 (C) 296 (D) 67

9. Given below are two statements :

Statement I: The function $f:\mathbf{R}\to\mathbf{R}$ defined by $f(x)$ $\frac{x}{1+|x|}$ is one-one.

Statement II: The function $f:\mathbf{R}\to\mathbf{R}$ defined by $f(x)$ $\frac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the correct swer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

10. An ellipse has its center at $(1,-2)$, one focus at $(3,$ and one vertex at $(5,-2)$. Then the length of its latus tum is :

(A) $\frac{16}{\sqrt{3}}$ (B) $4\sqrt{3}$

(C) 6 (D) $6\sqrt{3}$

11. Let $f(x)=\lim_{\theta\to 0}\left(\frac{\cos\pi x-x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right), x\in\mathbf{R}$. Cons the following two statements :

(I) $f(x)$ is discontinous at $x=1$.

(II) $f(x)$ is continous at $x=-1$.

Then,

(A) Only (I) is True (B) Neither (I) nor (II) is True

(C) Only (II) is True (D) Both (I) and (II) True

12. $\frac{6}{3^{26}}+\frac{10\cdot 1}{3^{25}}+\frac{10\cdot 2}{3^{24}}+\frac{10\cdot 2^2}{3^{23}}+\ldots+\frac{10\cdot 2^{24}}{3}$ is equal to:

(A) 2^{25} (B) 3^{25} (C) 3^{26} (D) 2^{26}

13. The sum of the coefficients of x^{499} and x^{500} in ($x)^{1000}+x(1+x)^{999}+x^2(1+x)^{998}+\ldots+x^{1000}$ is :

(A) ${}^{1000}\mathrm{C}_{501}$ (B) ${}^{1002}\mathrm{C}_{501}$

(C) ${}^{1001}\mathrm{C}_{501}$ (D) ${}^{1002}\mathrm{C}_{500}$

14. Let

$A=\{z\in\mathrm{C}:|z-2|\leqslant 4\}\, and$

$B=\{z\in\mathrm{C}:|z-2|+|z+2|=5\}$

Then the max $\{|z_1-z_2|: z_1\in$ A and $z_2\in$ B$\}$ is:

(A) $\frac{17}{2}$ (B) 8

(C) $\frac{15}{2}$ (D) 9

15. Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

(A) 41 (B) 89 (C) 80 (D) 32

16. Let the circle $x^2 + y^2 = 4$ intersect x-axis at the points $A(a, 0), a > 0$ and $B(b, 0)$. Let $P(2\cos\alpha, 2\sin a)$, $0 < \alpha < \frac{\pi}{2}$ and $Q(2\cos\beta, 2\sin\beta)$ be two points such that $(\alpha - \beta) = \frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on:

(A) $x^2 + y^2 - 4y - 4 = 0$ (B) $x^2 + y^2 - 4x - 4y = 0$

(C) $x^2 + y^2 - 4x - 4y - 4 = 0$ (D) $x^2 + y^2 - 4x - 4 = 0$

17. The sum of all the elements in the range of $f(x) = \operatorname{Sgn}(\sin x) + \operatorname{Sgn}(\cos x) + \operatorname{Sgn}(\tan x) + \operatorname{Sgn}(\cot x)$,
$x \neq \frac{n\pi}{2}, n \in Z$,
where $\operatorname{Sgn}(t) = \begin{cases} 1, & if t > 0 \\ -1, & if t < 0 \end{cases}$, is :

(A) 2 (B) 4 (C) -2 (D) 0

18. Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to :

(A) $-\frac{33}{56}$ (B) $\frac{16}{63}$

(C) $\frac{33}{56}$ (D) $-\frac{16}{63}$

19. Given below are two statements :
Statement I : $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7
Statement II : The integral part of $(7 + 4\sqrt{3})^{25}$ is an number.
In the light of the above statements, choose the cor answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

20. Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}$, $a > 2$. If such that $a, 4, \alpha, b$ are in A.P., then the equation αx $ax + 2(\alpha - 2b) = 0$ has :

(A) one root in $(0, 2)$ and another in $(-4, -2)$

(B) one root in $(1, 4)$ and another in $(-2, 0)$

(C) complex roots of magnitude less than 2

(D) both roots in the interval $(-2, 0)$

Math Sec 2

21. Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{(x-t)} f(t)dt, x \in \mathbf{R}$ and let $g(x) = \int_0^x (f(t) + 2)^{15}(t - 4)^6(t + 12)^{17}dt, x \in \mathbf{R}$. If pand qare respectively the points of local minima and local maxima of g, then the value of $|p + q|$ is equal to ____。

22. Three persons enter in a lift at the ground floor. The lift will go upto 10^{th} floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to

23. Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such $A^{100} = 100B + 1$. Then the sum of all the element B^{100} is ____

24. If $\sum_{r=1}^{25}\left(\frac{r}{r^4+r^2+1}\right) = \frac{p}{q}$, where p and q are positive inte such that $\gcd(p, q) = 1$, then $p + q$ is equal to ____。

25. If the distance of the point $P(43, \alpha, \beta), \beta < 0$, from line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k}), \mu \in \mathbf{R}$ along a line direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is e to ____

Phy Sec 1

26. For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

(A) $\sqrt{2} < n < 2\sqrt{2}$ (B) $\sqrt{2} < n < 2$

(C) $1 < n < 2$ (D) $n \geqslant 2$

27. Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is ____

(Given : $h = 6.63 \times 10^{-34} J \cdot s$ and $c = 3 \times 10^8 m/s$)

(A) 2×10^{16} (B) 10×10^{15}

(C) 5×10^{15} (D) 5×10^{16}